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Food Research International 39 (2006) 840–846
FOOD RESEARCH INTERNATIONAL
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# Combined effect of plasticizers and surfactants on the physical properties of starch based edible films

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Received 5 December 2005; accepted 6 April 2006

**Abstract**

Films made of potato starch were developed and glycerol as plasticizer and Tween 20, Span 80, and soy lecithin as surfactants were included in the formulation. Films were characterized with respect to water vapor permeability (WVP) and mechanical properties. The wettability of the film solutions was quantified by measuring their surface tension.

The incorporation of plasticizers resulted in more flexible and manageable films and higher WVP. At low concentration, Tween 20 was the surfactant that reduced surface tension the most, while at high concentration it was lecithin. In the absence of glycerol, surfactants had a significant effect on mechanical properties, but they did not modify significantly WVP of the films. It was observed a synergistic behavior between the plasticizer and the surfactants. Films with glycerol and high level of any of the surfactants behaved as films with larger amount of plasticizer (with lower tensile strength, higher elongation, and higher WVP). Tween 20 was the surfactant that showed the most intense synergistic effect with glycerol.
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*Keywords:* Starch; Edible film; Mechanical properties; Water vapor permeability; Surface tension

## 1. Introduction

The use of edible films and coatings has been constantly increasing in the food industry. Coatings help meet many challenges related to the storage and marketing of food products. The functionality and performance of edible films and coatings depend on their barrier and mechanical properties, which in turn depend on film composition, its formation process and the method of application onto the product.

Edible films include lipids, proteins, and carbohydrates such as cellulose, starch, and their derivates in their formulation. Starch is one of the most commonly used agricultural raw material, since it is a renewable source, inexpensive (even cheaper than polyethylene), widely available and relatively easy to handle. Starch is a polymeric carbohydrate composed of anhydroglucose units. This is not an uniform material and most starches contain two types of glucose polymers: a linear chain molecule termed amylose and a branched polymer of glucose termed amylopectin (Bertoft, 2004). The ratio amylose:amylopectine content is critical for the technological applications (Lourdin, Della Valle, & Colonna, 1995). The potential of starch as a material for edible films and for biomaterials has been widely recognized (Krochta & DeMulder-Johnson, 1997; Ropert, 1996). It is an appropriate matrix-forming material and it provides a good barrier to oxygen and carbon dioxide transmission, but a poor barrier to water vapor (Arvanitoyannis, Nakayama, & Aiba, 1998; Pagella, Spigno, & De Faveri, 2002). One of the most important limitation of the use of starch for films and coatings is the hydrophilicity of this material. Research has been conducted to improve the barrier properties of starch based films by adding lipids (García, Martino, & Zaritzky, 2000a).

The functional, organoleptic, nutritional, and mechanical properties of an edible film can be modified by the

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doi:10.1016/j.foodres.2006.04.002

addition of various chemicals in minor amounts. Plasticizers, such as glycerol, are often used to modify the mechanical properties of the film (Gaudin, Lourdin, Le Botlan, Ilari, & Colonna, 1999; Myllarinen, Partanen, Seppala, & Forssell, 2002). Plasticizers decrease intermolecular attractions between adjacent polymeric chains increasing film flexibility, but also they may cause significant changes in the barrier properties of the film (García, Martino, & Zaritzky, 2000b).

Surfactants could be incorporated into the film formulation to reduce surface tension of the solution, improving the wettability and adhesion of the film (Hiemenz & Rajagopalan, 1997). These chemicals are characterized for being amphiphilic. The lipophilic part of the molecule tends to be in a nonpolar environment and the hydrophilic part prefers to be in a polar (aqueous) environment (Stauffer, 1999). The balance between two, governs the functionality of the surfactant at interfaces and also its influence on the properties of the resulting films. This balance, called the hydrophilic/lipophilic balance (HLB), attaches a number to surfactants that guides its selection for a particular application. Thus, the higher HLB the better "oil in water" emulsifier and the lower HLB the better "water in oil" emulsifier. With surfactant addition to film formulation we expected a decrease in water vapor permeability values, because we were incorporating components with some hydrophobic characteristic.

Surface tension is roughly a linear function of surfactant concentration up to the critical micelle concentration (CMC), that is the concentration of a surfactant in aqueous solution at which colligative properties cease to change with increase in concentration. Then, above this limit, the thermodynamic activity of the surfactant does not increase with the addition of more surfactant, and the surface tension remains constant (Stauffer, 1999).

The purpose of the present work was to evaluate the influence of the combined presence of surfactants and plasticizers on the hydrophilicity and on the mechanical properties of starch films. These properties were quantified by water vapor permeability measurements and by a tensile test, respectively. The influence of these additives on the surface tension of the film solution was also measured.

## 2. Materials and methods

### 2.1. Materials

Potato starch used to make films was provided by AVEBE (Veendam, Netherlands). Glycerol was obtained from Panreac (Barcelona, Spain). Tween 20 (HLB 16.7), Span 80 (HLB 4.3) supplied by Merck (Hohenbrunn, Germany) and soy lecithin (HLB 8) from Jescuder (Barcelona, Spain) were employed as surfactants.

Sodium chloride was used to equilibrate films at 75% RH prior to tensile property testing, and silica gel from Merck (Darmstadt, Germany), to equilibrate the RH at 0% during WVP determination.

### 2.2. Film formation

Aqueous solutions of 2% (w/w) potato starch were prepared and heated at 70 °C for 30 min in a water bath. A weight of glycerol equal to 20% of the original starch was added and the solution was kept hot with mechanical stirring for five more minutes. Finally, surfactants were incorporated in a proportion of 0.5% or 5% relative to initial starch weight. This mixture was homogenized using an Ultra-Turrax T25 (Ika-Werke, Germany) for 10 min at 19,000 rpm, followed by 5 min at 22,000 rpm. Immediately after, a vacuum was applied to remove dissolved air.

Films were cast by pouring 50 g of solution onto a 14.7 cm internal diameter Petri dishes resting on a levelled surface. Fifteen films were prepared for each formulation. Samples were dried in a climatic chamber (Weiss Technik, Germany) with controlled conditions (60 °C and 60% RH) for approximately 24 h. Finally films were separated from the Petri dishes for their evaluation.

### 2.3. Film thickness measurement

Thickness of films was determined using a micrometer (Mitutoyo Corp., Code No. 543-551-1, Model ID-F125, Japan) to the nearest 0.001 mm at 10 random positions around the film, and average values were used in calculations.

### 2.4. Surface tension analysis

The ring method with a tensiometer K9 (Krüss, Alemania) was used to determine surface tension of the solutions. All measurements were made at 20 °C. Three replications were performed in all cases.

### 2.5. Water vapor permeability measurement

The gravimetric Modified Cup Method based on ASTM E96-92 (McHugh, Avena-Bustillos, & Krochta, 1993) was used to determine water vapor permeability (WVP). A cabinet (Sanplatec Corp., Japan) containing a fan (Elco, Milano, Italy) was used to test film WVP. Fan speeds were set to achieve an air velocity of 150 m/min to ensure uniform relative humidity throughout the cabinet. Prior to each experiment, cabinet was equilibrated to 0% RH using silica gel (Merck, Darmstadt, Germany).

Circular test cups were made of polymethylmethacrylate with 5.08 cm internal diameter and 1.1 cm height. Cup walls were sufficiently thick enough to make cups be impermeable to water vapor. Films without pinholes or any defects were sealed to the cup mouth with silicon (Rhodorsil Silicones, France) and a polymethylmetacrylate ring which was held with four screws symmetrically located around the cup circumference. Previously, 6 ml of deionized water was placed at the bottom of the test cups to expose the film to a high RH inside them. Average stagnant air gap heights between the water surface and the film were

determined. After assembly, the cups were placed in the 0% RH cabinet. Within 2 h, steady state had been achieved; five weights were then taken for each cup at 3 h intervals. Six replicates of each film were tested. Water vapor permeability of film was calculated as described by McHugh et al. (1993).

### 2.6. Mechanical properties

Tensile strength and percentage of elongation at break as evaluated in a tensile test was performed using a Texture Analyzer TA.XTZi (Surrey, United Kingdom). The films were cut in strips (25.4 × 90 mm) and conditioned at 75% RH for 7 days. Before the testing, thickness of the strips was measured in 10 points. Force and distance were recorded during extension of the strips at 0.8 mm/s up to break, and tensile strength and percentage elongation were calculated. At least 10 replications of each test sample were run.

### 2.7. Experimental design and statistical analyses

Experimental design was based on the variation of the following formulation parameters: glycerol content (0% or 20% with respect to starch content), type of surfactant (Tween 20, Span 80, or soy lecithin), surfactant content (0%, 0.5%, or 5% with respect to starch content).

SPSS version 12.0 was used for all statistical analyses. Analyses of variance (ANOVA) and Duncan means comparison test with a significance level of 0.05 were applied.

## 3. Results and discussion

### 3.1. Film characterization

Films obtained from 2% starch dispersions were translucent and they showed an homogeneous appearance, smooth surfaces without pores and cracks. It was possible to form stand-alone films without plasticizer addition, which indicated that starch is a good matrix-forming material. The incorporation of glycerol, increased starch films flexibility, making them easier to handle. The maximum compatible concentration of glycerol in the formulations was 20% (w/w) respect to starch content. When this limit was exceeded, phase separation occurred, leading to a physical exclusion of the plasticizer from the film, which corresponded to an extra adhesiveness of the films and an undesirable sticky aspect. Films containing surfactants needed the addition of glycerol for film integrity. It was not possible to obtain intact films with 5% surfactant or more, without plasticizer. The brittleness of films with surfactants but with no glycerol in their formulation was likely due to the lack of strong interactions between the surfactant and the biopolymer which could cause a phase separation during drying.

### 3.2. Surface tension

Preliminary experiments were conducted to study the influence of surfactant concentration on surface tension of starch solutions (Fig. 1). It was concluded for the three studied surfactants that minimum surface tension was reached with concentrations of 20% or above. Therefore, the CMC of those surfactants was likely to be around that value and thus the optimum surfactant concentration for our experiments. However, it was not possible to obtain intact films with surfactant concentration above 5%. Thus, we limited our study to three surfactant concentration values: 0%, 0.5%, and 5% w/w respect to starch content.

The presence of starch or glycerol did not modify significantly the surface tension of water (71.0 mN/m) . Thus, surface tension values for a 2% starch solution was 70.0 mN/m and for the same solution with 20% glycerol was 70.3 mN/m. These results were expected, since neither starch nor glycerol are tensio-active substances. The key additives were the surfactants, which produced modifications depending on type and concentration. Increasing amounts of surfactant led to lower surface tension (Fig. 2). The decrease of surface tension in film solution with low surfactant content (0.5%) was significant only with Tween 20.

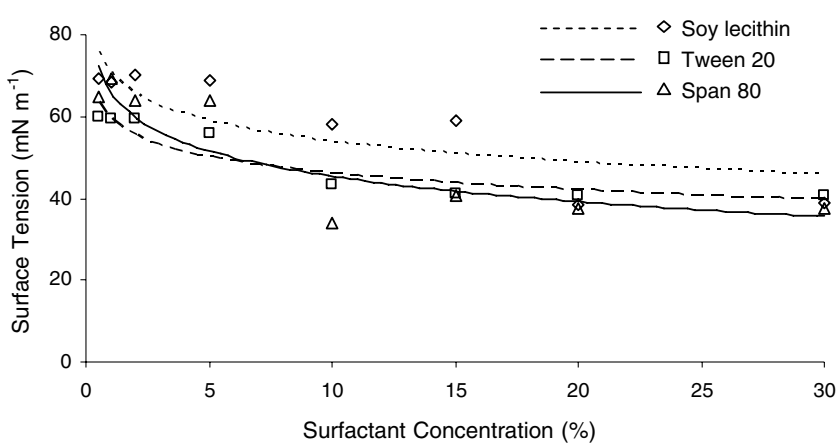


Fig. 1. Effect of surfactant concentration on surface tension of 2% starch solution without glycerol. Surfactant concentration is measured respect to starch weight.

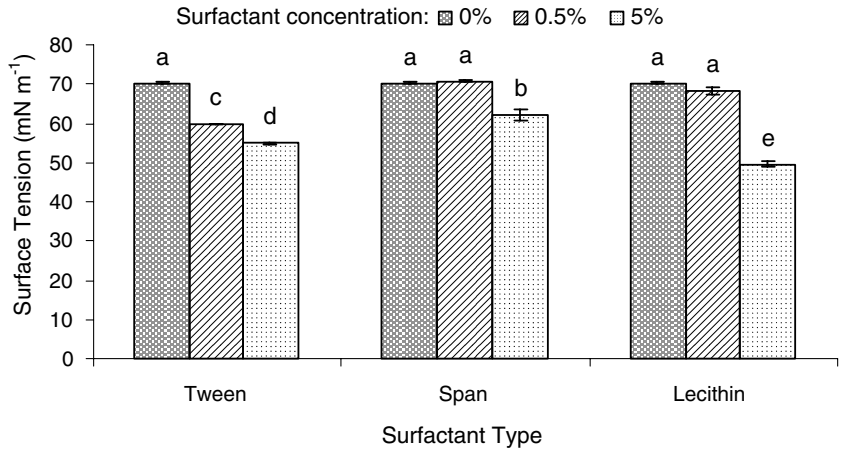


Fig. 2. Surface tension changes of film formulations with 20% glycerol with different types and concentrations of surfactants. Different letters denote significant differences ($P > 0.05$) through the Duncan test.

On the other hand, when we used high surfactant concentration (5%), significantly lower surface tension values were exhibited for all surfactants. As it can be observed in Fig. 2, lecithin was the surfactant that achieved the lowest value (49.6 mN/m). Results showed that Tween 20 was the surfactant with more uniform behavior, diminishing surface tension, even at low concentration, while Span 80 and soy lecithin needed higher amounts to be effective.

### 3.3. Mechanical properties

Results of the tensile tests are shown in Table 1. As expected, glycerol caused significant differences in both tensile strength and elongation. The plasticizer molecules interfered with starch packing, decreasing intermolecular attraction and increasing polymer mobility. This involved an increase in elongation and a decrease in tensile strength as glycerol content rose in film formulations. Similar results have been previously reported for hydroxypropyl starch films (Arvanitoyannis et al., 1998) and for potato and barley starch films (Koskinen, Suortti, Autio, Myllärinen, & Poutanen, 1996). In the case of the starch films without surfactant, the addition of glycerol diminished tensile strength from 0.44% to 0.20 MPa and the maximum elongation changed from 5.9% to 12.1%.

If there is no glycerol in the formulation, the presence of any of the surfactants (studied only at the lower level) caused a slight reduction of tensile strength (only significant for lecithin) and a reduction of elongation (significant for span and lecithin). In other words, films without plasticizers and with surfactants are slightly weaker and less flexible, and this is more intense and significant with Span 80.

If there is glycerol in the formulation, the addition of surfactants at both studied concentration caused an important and significant reduction of tensile strength (except for Tween at 0.5%). The effect of the surfactant on the elongation of films with glycerol depended on surfactant concentration. Thus, the addition of lower amount of surfactant reduced film elongation. However, when high content of surfactant was used, an important and a significant increase in the elongation was found, regardless surfactant type. That is, films with glycerol and with high content of surfactant unexpectedly were found to have significantly

Table 1
Tensile strength and elongation at break in a tensile test for starch based films*

| Glycerol (%) | Surfactant (%) | Thickness (μm) | Tensile strength (MPa) | Elongation at break (%) |
|---|---|---|---|---|
| 0 | 0 | 50 ± 5 | 44.1 ± 2.7 a | 5.9 ± 1.1 a |
| | 0.5 Tween | 53 ± 5 | 42.1 ± 4.4 a | 5.4 ± 1.0 a |
| | 0.5 Span | 49 ± 4 | 40.6 ± 5.4 ab | 4.2 ± 0.8 b |
| | 0.5 Lecithin | 54 ± 9 | 37.2 ± 3.9 b | 3.7 ± 0.4 b |
| 20 | 0 | 60 ± 5 | 20.3 ± 4.9 B | 12.1 ± 6.8 C |
| | 0.5 Tween | 50 ± 7 | 23.4 ± 3.7 A | 5.2 ± 1.0 D |
| | 0.5 Span | 64 ± 3 | 14.2 ± 3.3 C | 7.9 ± 1.1 CD |
| | 0.5 Lecithin | 55 ± 12 | 16.1 ± 4.4 C | 8.0 ± 2.7 CD |
| | 5 Tween | 78 ± 24 | 2.4 ± 0.3 E | 31.5 ± 5.7 A |
| | 5 Span | 89 ± 16 | 3.2 ± 0.5 E | 27.6 ± 5.5 A |
| | 5 Lecithin | 72 ± 12 | 7.3 ± 1.6 D | 21.1 ± 9.5 B |

* Two means followed by the same letter in the same column (different analysis for different amounts of glycerol) are not significantly ($P > 0.05$) different through the Duncan test.

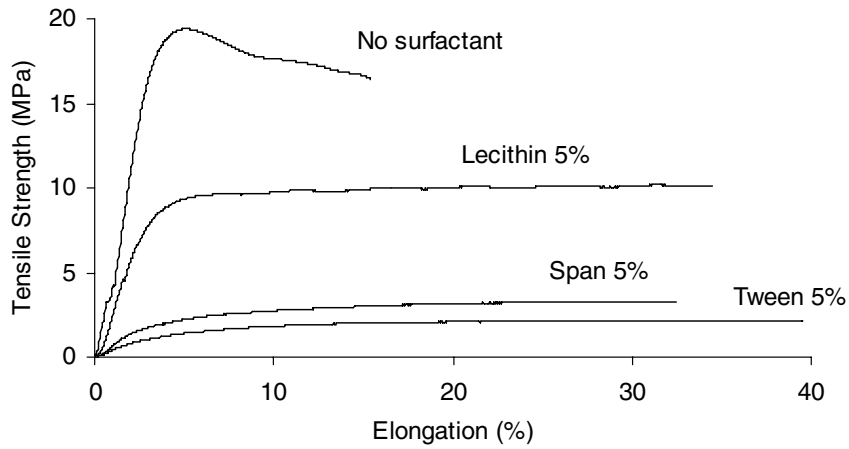


Fig. 3. Effect of the presence of surfactants on the mechanical properties of starch films with 20% glycerol.

lower tensile strength and higher elongation than films without surfactant.

Therefore, there is an apparent synergistic effect between glycerol and the studied surfactants at high concentration. This interaction meant that films with glycerol in the presence of high level of surfactant behaved mechanically as films with larger amount of plastizicers (with lower tensile strength and higher elongation) (Fig. 3). This apparent plastic behavior of films with high content of surfactant only happened in the presence of glycerol (films with high level of surfactant and without glycerol were too brittle to be analyzed).

A possible explanation of the previous phenomenon is that surfactants are small size molecules that could stay between starch chains, just like glycerol, increasing even more chain mobility and enhancing the initial plastic effect. The hydrophilic part of the surfactant could interact with glycerol or water facilitating its presence between starch chains. Tween 20 was the surfactant that showed the most extreme behavior, with a tensile strength of only 2.4 MPa and an elongation at break of 31.5%. These values meant a reduction of 90% of the tensile strength and an increase of about 250% of the elongation in relation to films with just glycerol (Table 1). A possible reason of the previous observation is that Tween 20 was the most hydrophilic of the studied surfactants (highest HLB value), and then, the one with more theoretical chances to interact with glycerol or water with the consequences explained above.

### 3.4. Water vapor permeability (WVP)

The presence of plasticizers increased water vapor transmission of films regardless of formulation, and as a consequence, permeability values were always significantly higher for plasticized films, as it can be observed in Fig. 4. As pointed out earlier, glycerol is a relatively small hydrophilic molecule, which can be inserted between adjacent polymeric chains, decreasing intermolecular attractions and increasing molecular mobility, which facilitates the migration of water vapor molecules. Similar results have been reported for wheat gluten (Gontard, Guilbert, & Cuq, 1993), sodium caseinate and soluble starch (Arvanitoyannis & Biliaderis, 1998) and whey protein (McHugh & Krochta, 1994; Sothornvit & Krochta, 2000). More-

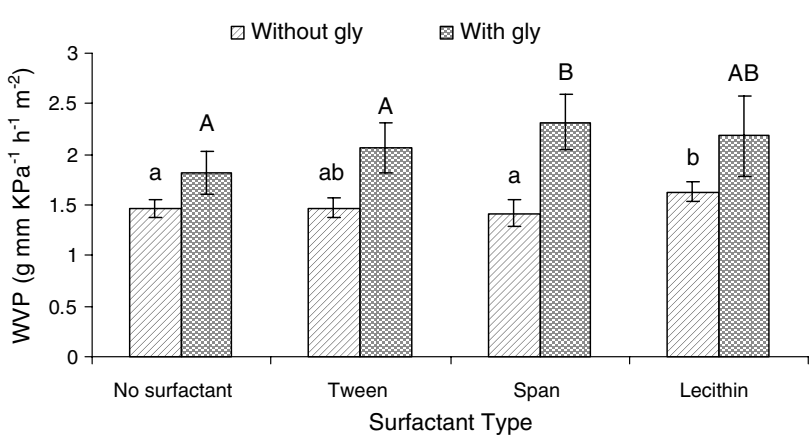


Fig. 4. Effect of presence of 20% glycerol on WVP of starch films with 0.5% surfactant. Different letters (a–b or A–B) denote significant differences ($P > 0.05$) through the Duncan test.

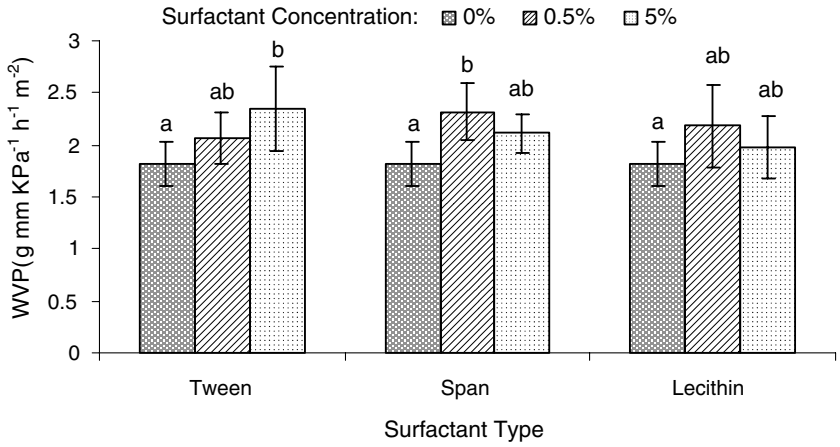


Fig. 5. WVP of starch based films with 20% glycerol and different types and concentrations of surfactants. Different letters denote significant differences ($P > 0.05$) through the Duncan test.

over, this hydrophilic plasticizer is compatible with an hydrophilic film forming material, such as starch, and enhances sorption of polar molecules, such as water. With the addition of surfactants, we were incorporating components with some hydrophobic characteristics in film formulations, so we expected a decrease in water vapor permeability. However, low level of surfactant in the absence of glycerol did not modify significantly this property regardless surfactant type, and we obtained very similar values to those for film based just on starch (Fig. 4).

Results were different when we studied the effect of both additives at the same time, that is, plasticizers and surfactants (Fig. 5). As we observed for mechanical properties, an apparent synergistic effect between both components is shown, appreciable for low and high level of surfactants. The effect of glycerol on WVP was enhanced with the incorporation of surfactants, increasing its value regardless of surfactant type and concentration. We expected just the opposite due to the partially hydrophobic character of surfactants.

It can be observed a different trend for Tween 20 as opposed of that for Span 80 and lecithin (Fig. 5). Tween 20 presented a linear trend, increasing WVP as surfactant concentration increased. However the addition of higher contents of Span 80 and lecithin led to lower WVP, but not in a significant way. These results could be explained by the HLB ratio. Tween 20, with an HLB of 16.7, was the most hydrophilic surfactant, which gave higher permeability values. Span 80 and lecithin, with HLB of 4.3 and 8, respectively, had more hydrophobic character, so increasing their presence, water vapor transmission diminished.

## 4. Conclusions

Starch is a good matrix-forming material that allows the elaboration of edible films even without plasticizer with the stability and cohesion needed for their surface, permeability, and mechanical characterization. Glycerol is a plasticizer compatible with starch, improving film flexibility, facilitating its handling and preventing cracks. The presence of the plasticizer led to greater mobility of the polymer matrix as shown by the lower tensile strength and higher elongation and WVP values.

Surfactants improve wettability properties of film solutions by decreasing surface tension. We found Tween 20 to be the most effective surfactant at low concentration and soy lecithin to decrease surface tension the most with high level. A synergistic behavior between glycerol and surfactants was reflected in both mechanical and mass transfer properties. Thus, the presence of surfactants enhanced the plasticity and increased the WVP of films with glycerol. This likely showed some interaction between the glycerol and the surfactants that allow more molecular mobility. This phenomenon should be the focus of future studies.

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